

ZACATECAS INTL, MexicoWMO: **765255**Lat: **22.897N**Lon: **102.687W**Elev: **2177**StdP: **77.77**Time Zone: **-6.00 (MEX)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.4	-0.2	-16.1	1.2	5.7	-14.2	1.4	7.9	15.1	17.5	13.3	16.2	1.8	180	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	17.5	30.8	13.0	29.1	12.4	28.1	12.3	16.2	22.8	15.7	22.6	15.4	22.4	5.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	13.1	18.3	13.2	12.4	18.0	13.0	12.2	18.0	54.1	23.0	52.4	22.8	51.0	22.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	10.4	9.7	-7.0	32.3	3.0	1.1	-9.2	33.1	-10.9	33.8	-12.6	34.4	-14.8	35.2		
(5)			-8.8	18.0	2.2	1.5	-10.4	19.1	-11.7	20.0	-13.0	20.9	-14.6	22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	10.8	12.7	14.5	17.4	19.7	20.1	18.8	18.6	17.7	15.9	13.1	11.1
	DBStd	3.79	2.28	2.43	2.39	2.06	2.04	1.72	1.38	1.32	1.42	1.99	2.47	2.40
	HDD10.0	47	19	6	2	0	0	0	0	0	0	0	5	15
	HDD18.3	1077	234	157	121	40	10	4	12	12	27	79	158	223
	CDD10.0	2192	43	83	140	223	301	303	271	267	231	182	97	51
	CDD18.3	179	0	0	1	13	52	58	24	21	8	2	0	0
	CDH23.3	3291	4	55	202	619	969	665	265	246	111	112	37	6
CDH26.7	674	0	2	18	129	295	177	23	20	6	4	1	0	
Wind	WSAvg	3.9	4.1	4.7	4.9	4.9	4.4	3.9	3.6	3.2	2.9	3.2	3.5	4.0
Precipitation	PrecAvg	445	15	12	5	6	16	70	93	87	91	30	9	6
	PrecMax	668	139	74	67	34	52	194	299	180	168	89	44	46
	PrecMin	201	0	0	0	0	0	18	20	20	26	0	0	0
	PrecStd	131	26	18	15	9	11	47	65	41	35	24	10	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	26.9	28.2	30.9	32.2	31.9	28.2	28.2	27.2	27.1	26.1	24.1
		MCWB	9.6	10.4	11.1	12.6	13.3	13.4	13.5	14.1	14.0	13.0	11.8	10.3
	2%	DB	22.8	25.1	27.0	29.1	31.0	30.1	27.1	27.1	26.0	26.0	24.2	22.9
		MCWB	9.1	9.9	10.4	11.8	13.0	13.2	13.6	14.1	14.0	12.9	11.3	9.6
	5%	DB	21.2	23.9	25.9	28.1	29.5	28.8	26.1	26.0	24.2	24.2	23.1	21.8
		MCWB	8.5	9.4	10.1	11.3	12.3	13.2	13.8	14.3	14.3	12.9	10.9	9.3
	10%	DB	19.9	22.2	24.1	27.0	28.2	27.2	24.9	24.7	23.2	23.1	22.0	20.2
		MCWB	8.0	9.0	9.5	10.7	11.9	13.3	13.9	14.1	14.5	12.8	10.5	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.1	12.2	13.4	15.3	16.5	16.2	16.7	17.1	16.2	14.5	12.3
		MCDB	20.5	22.8	23.9	27.1	26.1	24.4	22.8	22.8	22.7	22.3	22.2	19.5
	2%	WB	10.3	10.9	11.4	12.5	14.2	15.7	15.6	16.0	16.2	15.4	13.4	11.1
		MCDB	19.6	22.4	23.8	26.9	25.6	23.7	22.6	22.7	22.2	22.1	21.2	20.1
	5%	WB	9.6	10.2	10.6	11.8	13.6	15.1	15.1	15.5	15.7	14.6	12.4	10.2
		MCDB	18.9	21.9	23.6	26.6	25.6	23.5	22.5	22.5	21.7	21.6	20.0	19.4
	10%	WB	8.7	9.5	10.0	11.2	12.9	14.6	14.6	14.9	15.2	13.9	11.5	9.5
		MCDB	17.8	21.0	23.1	25.7	25.5	23.0	22.0	22.2	21.2	20.8	19.3	18.4
Mean Daily Temperature Range	5% DB	MDBR	18.1	18.4	19.2	19.4	17.5	13.9	12.3	12.7	12.1	15.4	17.8	18.0
		MCDBR	20.5	21.2	21.2	20.6	19.2	17.0	14.6	15.0	14.4	17.7	19.5	20.4
		MCWBR	10.5	10.3	10.0	9.1	7.7	6.1	5.2	5.6	6.1	8.1	9.4	10.4
	5% WB	MCDBR	16.8	18.4	18.9	19.3	16.0	12.6	11.6	11.9	11.1	13.7	15.3	17.3
		MCWBR	9.0	9.1	9.1	8.7	6.1	4.7	4.8	5.1	5.2	6.6	7.8	9.1
Clear-Sky Solar Irradiance	taub	0.265	0.276	0.278	0.299	0.354	0.348	0.352	0.344	0.345	0.310	0.274	0.272	
	taud	2.550	2.508	2.503	2.422	2.325	2.381	2.411	2.445	2.435	2.524	2.594	2.568	
	Ebn at Noon	1016	1020	1027	1001	936	934	931	943	942	969	999	996	
	Edh at Noon	91	102	107	119	130	122	119	115	114	99	87	87	
All-Sky Solar Radiation	RadAvg	4.68	5.66	6.75	7.38	7.31	6.71	6.42	6.33	5.54	5.48	4.96	4.43	
	RadStd	0.33	0.49	0.34	0.41	0.38	0.44	0.46	0.34	0.40	0.35	0.38	0.28	